

3

Friction

Dry Friction

When two dry surfaces in contact tend to move over each other, a resistance is set up. This resistance is called force of dry friction. The direction of this frictional force is parallel to the surfaces in contact and opposite to the direction in which the body tends to move with respect to the other.

The cause of dry friction is the interlocking of microscopic projections on the surfaces.

Let two bodies A and B have relative motion with respect to each other as shown in fig (a).



Figure (a)



Figure (b): FreeBody Diagram of A

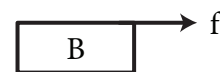


Figure (c): FreeBody Diagram of B

Fig (b) shows the free body diagram of body A. As the tendency of A is to move towards right with respect to B, the frictional force 'f' will be in opposite directions, is, towards left.

Fig (c) show the free body diagram of body B. As the relative motion of B with respect to A is towards left, friction 'f' will be towards right.

Characteristics of Dry friction

Coefficient of friction (μ):-

Force of limiting friction f is directly proportional to the normal reaction R_N

$$f \propto R_N$$

$$f = \mu R_N$$

Where μ is called this coefficient of friction.

Angle of friction (ϕ):-

Ref figure ...

The resultant of f and R_N is denoted by R . The angle between R and R_N is called the angle of friction (ϕ).

The cone formed by R_N as the axis and R as generator is called the cone of friction, of which ϕ is the semi-cone angle.

$$\tan \phi = \frac{f}{R_N} = \frac{\mu R_N}{R_N} = \mu$$

$$\text{ie, } \mu = \tan \phi$$

Angle of Repose:-

It is the maximum angle to which is plane can be inclined so that a body can remain in equilibrium on it solely due to the friction between contact surfaces.

In such a case all the forces involved will keep the body in equilibrium.

$$R_N = mg \cos \alpha \quad (1)$$

$$f = \mu R_N = mg \sin \alpha \quad (2)$$

From (1) and (2),

$$\tan \alpha = \mu$$

But we know that $\mu = \tan \phi$

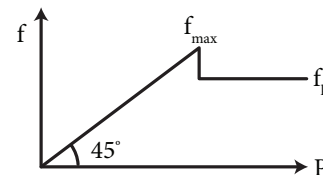
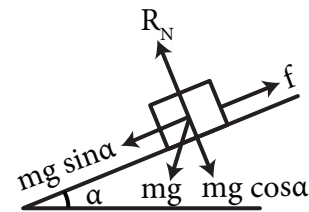
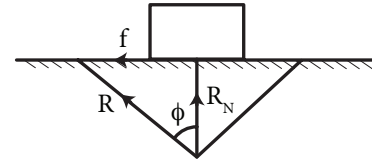
Hence, $\tan \alpha = \tan \phi$

or, $\alpha = \phi$

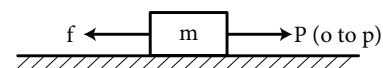
Therefore angle of repose is same as angle of friction.

Laws of Dry Friction:-

- 1) Frictional force will always be opposite to the direction in which the body moves or tends to move.
- 2) The magnitude of frictional force just suffice to prevent the body from moving
- 3) The limiting force of friction bears a constant ratio with the normal reaction. This depends on the roughen and other characterization of the surfaces in contact.
- 4) When the body starts to move, the magnitude of frictional resistance will be less than that offered during limiting equilibrium.



When a gradually increasing force is applied to a body, the frictional resistance increases proportionately to a maximum value ($f_{\max}$) and then drops. While the body moves, the frictional resistance will be f_k ($f_k < f_{\max}$).



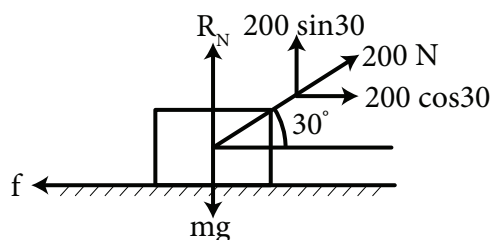
Problems 1

A pull of 200 N inclined at 30° to the horizontal just moves a body resting on a horizontal plane. Similarly a push of 250 N inclined at 30° to this horizontal is required to just move the body. Determine the weight of the body as well as the coefficient of friction.

Solution

Let the weight of the body be mg and the coefficient of friction be μ .

Case 1



For, vertical equilibrium,

$$200 \sin 30 + R_N = mg$$

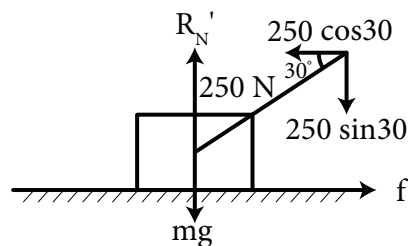
$$\text{ie, } 100 + R_N = mg \quad (i)$$

For horizontal equilibrium,

$$200 \cos 30 = f = \mu R_N$$

$$\text{ie, } 173.2 = \mu R_N \quad (ii)$$

Case 2



For vertical equilibrium,

$$R'_N = 250 \sin 30 + mg$$

$$\text{ie, } R'_N = 125 + mg \quad (iii)$$

For horizontal equilibrium,

$$250 \cos 30 = f = \mu R'_N$$

$$\text{ie, } 216.5 = \mu R_N' \quad (\text{iv})$$

Substitute $R_N' = mg - 100$ from (i) in equation (ii)

$$173.2 = \mu (mg - 100) \quad (\text{v})$$

Substitute $R_N' = 125 + mg$ from (iii) in equation (iv)

$$216.5 = \mu (125 + mg) \quad (\text{vi})$$

Divide (v) with (vi)

$$\frac{173.2}{216.5} = \frac{mg - 100}{125 + mg}$$

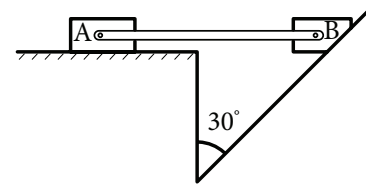
Solving, $mg = 1000 \text{ N}$

From eqn (v),

$$\mu = \frac{173.2}{(mg - 100)} = \frac{173.2}{900} = 0.192$$

Problem 2

Block B weighs 1000 N. The two blocks are connected using an inflexible rod as shown. Find out the minimum weight required for block A to keep the system in equilibrium (Assume $\mu = 0.2$) Assume that the inclined surface is frictionless.



Solution

Let the push exerted by the rod as the two bodies be p .

Free Body Diagram of block B

$$R_N \sin 30 = m_1 g \quad (1)$$

$$P = R_N \cos 30 \quad (2)$$

$$(1) \text{ divided by } (2) \text{ gives, } \tan 30 = \frac{m_1 g}{P}$$

$$\text{or } P = \frac{m_1 g}{\tan 30} = \frac{1000}{0.577} = 1733.1 \text{ N}$$

Free Body Diagram of block A

$$R_N' = m_2 g$$

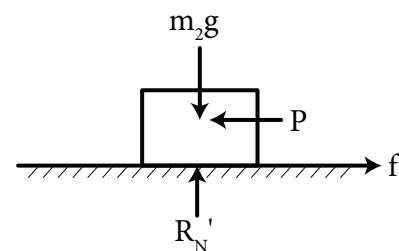
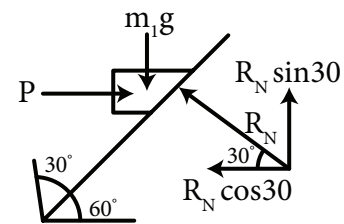
$$f = \mu R_N'$$

$$P = f$$

But $P = 1733.1 \text{ N}$

So, $f = 1733.1 \text{ N} = \mu R_N'$

$$R_N' \frac{1733.1}{\mu} = \frac{1733.1}{0.2} = 8665.5 \text{ N}$$



$$\text{ie, } m_2 g = R_N' = 8665.5 \text{ N}$$

Minimum weight of block A = 8665.5 N

Problem 3

Find the minimum force F required to move the block B if A weight 500 N and B weight 1000 N. Assume $\mu = 0.25$ for all contact surfaces.

Solution

Free Body Diagram of B

In addition to reaction from the ground and associated friction (R_1 and f_1), there is the reaction from the block A (R_2) and associated friction (f_2).

For horizontal equilibrium,

$$f_1 + f_2 = F$$

$$\text{or, } \mu R_1 + \mu R_2 = F \quad (\text{i})$$

For vertical equilibrium,

$$R_1 = R_2 + 1000 \quad (\text{ii})$$

Free Body Diagram of A

Here, T is the tension in the string. R_2 and f_2 will act in a direction opposite to that in the block B. (Newton's 3rd law).

For horizontal equilibrium,

$$T = f_2 = \mu R_2 \quad (\text{iii})$$

For vertical equilibrium,

$$R_2 = 500 \text{ N} \quad (\text{iv})$$

Substitute the value of R_2 from (iv) in Equation (i) and (ii)

$$R_1 = 500 + 1000 = 1500 \text{ N}$$

$$F = \mu(R_1 + R_2) = 0.25(1500 + 500)$$

$$= 500 \text{ N}$$

Problem 4

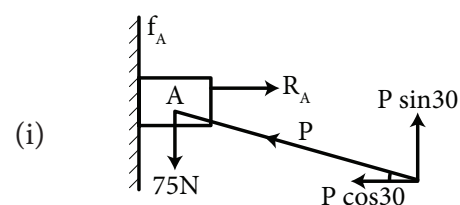
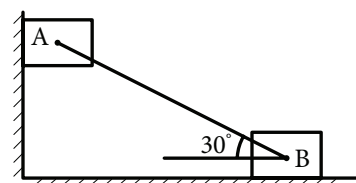
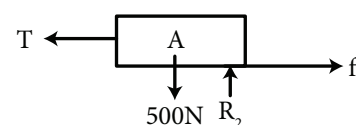
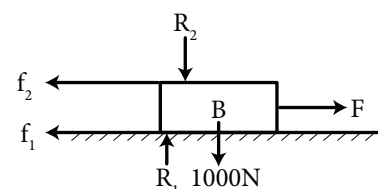
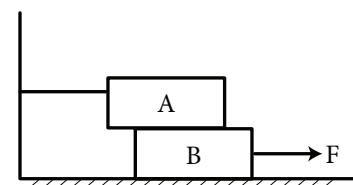
Two blocks are connected by a right bar at 30° inclination as shown. If the system is in equilibrium, determine the coefficient of friction ' μ ' which is same for the wall and the floor. Take $W_A = 75 \text{ N}$ and $W_B = 100 \text{ N}$

Solution

Free Body Diagram of A

$$P \cos 30 = R_A$$

$$P \sin 30 + f_A = 75$$



$$\text{ie, } P \sin 30 + \mu R_A = 75 \quad (\text{ii})$$

Free Body Diagram of B

$$P \sin 30 + 100 = R_B \quad (\text{iii})$$

$$P \cos 30 = f_B$$

$$\text{ie, } P \cos 30 = \mu R_B \quad (\text{iv})$$

From (i) and (ii)

$$P \sin 30 + \mu P \cos 30 = 75 \quad (\text{v})$$

From (iii) and (iv)

$$P \cos 30 = \mu (P \sin 30 + 100)$$

$$\text{ie, } P \cos 30 - P \mu \sin 30 = 100 \mu \quad (\text{vi})$$

(vi) divided by (v) given,

$$\frac{\cos 30 - \mu \sin 30}{\sin 30 + \mu \cos 30} = \frac{100 \mu}{75}$$

$$64.95 - 37.5 \mu = 50 \mu + \mu^2 (86.6)$$

$$86.6 \mu^2 + 87.5 \mu - 64.95 = 0$$

Solving $\mu = 0.497, -1.51$

So, coefficient of friction, $\mu = 0.497$

Problem 5

A ladder of length 20m and weight W is at an angle of 45° with the floor and supported by floor and wall. A man of weight $\frac{W}{2}$ climbs up the ladder. How far can be climb before the ladder slips? (Take $\mu = 0.33$)

Solution

Weight of the ladder acts at the midpoint, ie, at 10m.

Let the distance climbed by man before the ladder slips be 'x' m from the base, as shown.

When the ladder begins to slip, the base tends to move right, causing the friction f_1 , to act towards left.

Also the top tends to slip down, causing the friction to act upwards as shown.

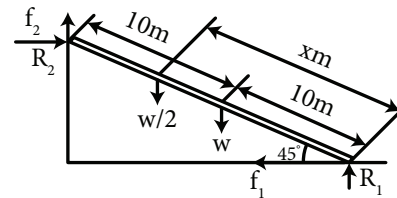
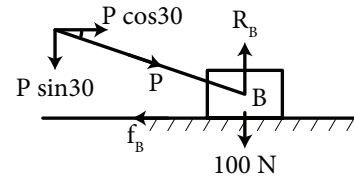
For vertical equilibrium,

$$f_2 + R_1 = \frac{W}{2} + W$$

$$\text{or, } \mu R_2 + R_1 = \frac{3}{2} W \quad (\text{i})$$

For horizontal equilibrium,

$$R_2 = f_1 \text{ or } R_2 = \mu R_1 \quad (\text{ii})$$



Taking moments about the base,

$$W(10 \cos 45) + \frac{W}{2}(x \cos 45) - f_2(20 \cos 45)$$

$$-R_2(20 \sin 45) = 0$$

$$(7.07)W + x(0.354)W - (0.33)R_2(14.14)$$

$$-R_2(14.14) = 0$$

$$(7.07)W + (0.354)xW - (18.81)R_2 = 0 \quad (\text{iii})$$

From (i) and (ii)

$$R_2 = \mu \left(\frac{3}{2}W - \mu R_2 \right) = 0.495W - 0.109R_2$$

$$1.109R_2 = 0.495W$$

$$\text{or, } R_2 = 0.446W \quad (\text{iv})$$

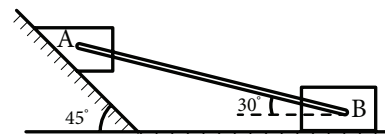
Substitute for R_2 from (iv) in eqn (iii)

$$(7.07) + (0.354)x - (18.81)(0.446) = 0$$

$$\text{or, } x = \frac{1.319}{0.354} = 3.73m$$

Problem 6

Block A and B weighing 1000 N and 2500 N are connected using a rigid bar as shown. What horizontal force F should be applied to block B in order to just prevent the block A from sliding down. Assume coefficient of friction for all surfaces as 0.25.



Solution

Let P be the push from the rod.

Resolve P along the inclined plane and perpendicular to it.

The angle between P and the inclined plane is $45 - 30 = 15^\circ$

Also resolve the weight 1000 N

Free Body Diagram of block A

For equilibrium along the incline,

$$P \cos 15 + f_1 = 1000 \sin 45$$

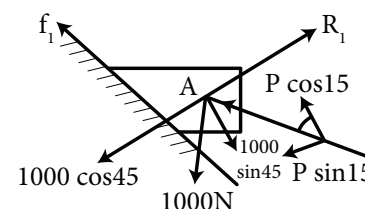
$$\text{or, } P \cos 15 + \mu R_1 = 1000 \sin 45 \quad (1)$$

perpendicular to the incline,

$$P \sin 15 + 1000 \cos 45 = R_1 \quad (2)$$

Substitute (2) in (1)

$$P \cos 15 + \mu(P \sin 15 + 1000 \cos 45) = 1000 \sin 45$$



$$(0.966)P + (0.065)P + 176.8 = 707.1$$

$$(1.031)P = 530.3$$

or, $P = 514.4 \text{ N}$

The same push p will be exerted on block B by the rod.

For Vertical equilibrium,

$$R_2 = P \sin 30 + 2500$$

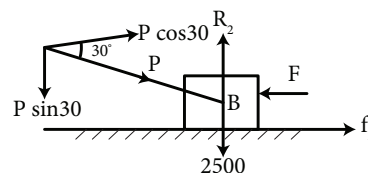
$$= (514.4)(0.5) + 2500 = 2757.2 \text{ N}$$

For horizontal equilibrium,

$$F = P \cos 30 + f_2$$

$$= (514.4)(0.866) + (0.25)(2757.2)$$

$$= 1134.7 \text{ N}$$



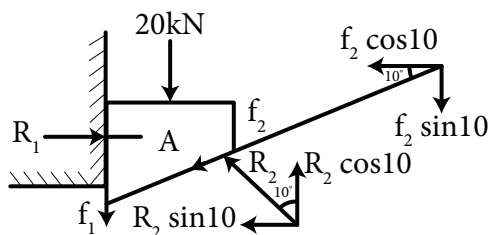
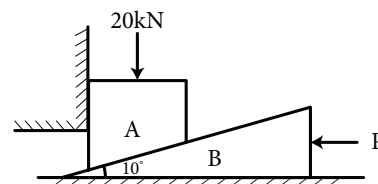
Problem 7

Refer figure...

Determine the magnitude of the horizontal force p to be applied to the weightless wedge B in order to raise the load A ($= 20 \text{ kN}$)

Assume $\mu = 0.2$ for all contact surfaces.

Solution



Free Body Diagram of block A

For horizontal equilibrium,

$$R_1 = R_2 \sin 10 + f_2 \cos 10$$

$$= R_2 \sin 10 + \mu R_2 \cos 10$$

$$\text{ie, } R_1 = (0.174)R_2 + (0.197)R_2$$

$$\text{or, } R_1 = 0.371R_2$$

(1)

For Vertical equilibrium,

$$R_2 \cos 10 = 20 + f_1 + f_2 \sin 10$$

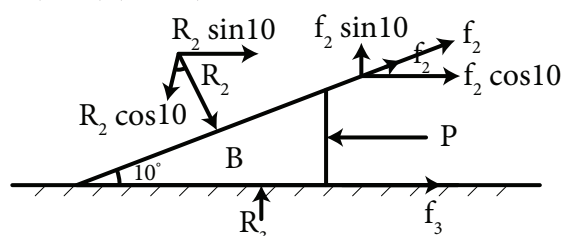
$$= 20 + (0.2)R_1 + (0.2)R_2 \sin 10$$

$$\begin{aligned} \text{or, } (0.985)R_2 &= 20 + (0.2)R_1 + (0.035)R_2 \\ (0.95)R_2 &= 20 + (0.2)R_1 \end{aligned} \quad (2)$$

From (1) and (2)

$$\begin{aligned} 0.95R_2 &= 20 + (0.2)(0.371)R_2 \\ 0.876R_2 &= 20 \\ R_2 &= 22.83 \text{ kN} \end{aligned}$$

$$\text{and } f_2 = (0.25)(22.83) = 5.71 \text{ kN}$$



Free Body Diagram of B

Here magnitudes of R_2 and f_2 will be same but direction will be opposite (Newton's 3rd law).

For Vertical equilibrium,

$$\begin{aligned} R_2 \cos 10 &= f_2 \sin 10 + R_3 \\ &= \mu R_2 \sin 10 + R_3 \end{aligned}$$

$$\text{ie, } (22.83)(0.985) = (0.2)(22.83)(0.174) + R_3$$

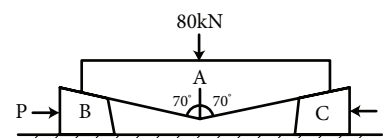
$$\text{or, } R_3 = 21.69 \text{ kN}$$

For horizontal equilibrium,

$$\begin{aligned} P &= f_3 + f_2 \cos 10 + R_2 \sin 10 \\ &= (0.2)(21.69) + (5.71)(0.985) + (22.83)(0.174) \\ &= 13.93 \text{ kN} \end{aligned}$$

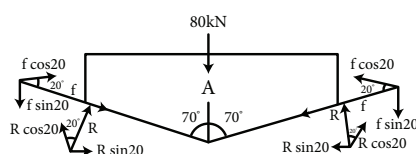
Problem 8

Two wedges B and C are used to raise the weight A ($w = 80 \text{ kN}$) as shown. Determine the magnitude of P assuming weightless wedges and μ as 0.25 for all contact surfaces.



Solution

Free Body Diagram of A



For horizontal equilibrium,

$$f \cos 20 + R \sin 20 = R \sin 20 + f \cos 20$$

Which mean forces cancel out

For vertical equilibrium

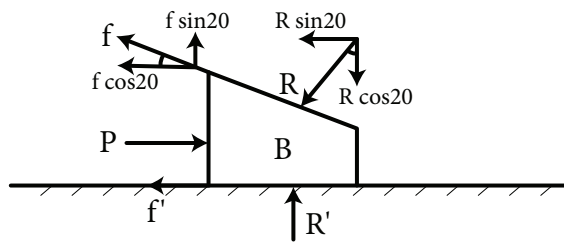
$$2(R \cos 20) = 2(f \sin 20) + 80$$

Put $f = (0.25)R$

$$(1.88)R = (0.171)R + 80$$

$$R = 46.8 \text{ kN}$$

Now Consider **Free Body Diagram of B**



R and f will be same in magnitude but direction will be opposite here.

For vertical equilibrium,

$$R \cos 20 = R' + f \sin 20$$

$$R' = R \cos 20 - \mu R \sin 20$$

$$= (46.8)(0.94) - (0.25)(46.8)(0.342)$$

$$= 40 \text{ kN}$$

For horizontal equilibrium,

$$P = R \sin 20 + f \cos 20 + f'$$

$$= (46.8)(0.342) + (0.25)(46.8)(0.94) + (0.25)(40)$$

$$= 37 \text{ kN}$$